Dry etching of $LiNbO_3$ using inductively coupled Plasma

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***Abstract*-In this letter, we report ridge waveguides fabricated on X-cut proton exchange (PE)-$LiNbO_3$ using inductively coupled plasma (ICP) etching techniques. Various etching masks and fluorine gases are investigated. Smooth etched surfaces are obtained by using Cr as a mask combined with SF_6/Ar etching gases. A high etch rate of 97.5 nm/min is achieved by using CHF_3/Ar gases. Ridge waveguides with approximately 600nm depth, smooth surfaces and nearly vertical sidewalls are successfully fabricated using optimized etching conditions.**

I. INTRODUCTION

Lithium niobate ($LiNbO_3$) is a nonlinear crystal material exhibiting properties such as piezoelectric, ferroelectric and the Pockels effects. It is a commonly-used material in the fabrication of nonlinear optical devices because of its large electro–optic coefficients [1], large transparency range and wide intrinsic bandwidth. Waveguides in $LiNbO_3$ crystals are traditionally fabricated by metal diffusion or annealed proton exchange (APE) processes [2]. Both techniques will induce an index change to the $LiNbO_3$ substrate and allow creation of slab or channel waveguides. But it is not suitable for fabricating ridge waveguides, which can provide better guiding in the lateral direction and increase the bandwidth of $LiNbO_3$ optical modulators [3]. So far several techniques have been developed for ridge waveguide fabrication, such as wet etching by HF based etchant [4], reactive ion etching (RIE) [5,6], and ion-beam enhanced etching(IBEE) [7]. Recently plasma dry etching has drawn increased attention because of controllable and highly anisotropic properties [8,9], but it is challenging because of the lack of a suitable mask and very low etch rates. Besides, it is hard for plasma etching to lead to a smooth surface, which is important for the reduction of optical loss. The deterioration of the surface after etching is caused by different mechanisms, like physical sputtering of the mask during etching and redeposition of LiF and NbF_x. Proton exchange (PE) technique can reduce the concentration of Li ions in the $LiNbO_3$ surface, and lead to reduction of redeposition of LiF and improve the etching rate. Consequently, plasma etching we studied was performed on PE $LiNbO_3$ samples.

In this work, the plasma etching processes using different metal masks and gas mixtures was investigated. Three metallic masks, silver (Ag), nickel (Ni), Chromium (Cr) were used for ICP etching. The etching parameters and gas mixture were optimized to avoid the surface roughness and enhance the etching rates. After etching, the surface condition, etching rate and profile were characterized by surface profile and scanning electron microscopy (SEM). Ridge waveguides with nearly vertical sidewalls and smooth surfaces were fabricated.

II. EXPERIMENTS

In our work, benzoic acid was chosen for PE process because it is less toxic and requires a lower temperature for exchange. The ridge waveguide fabrication process is depicted schematically in Fig. 1. The samples used in our experiments were X-cut $LiNbO_3$ wafers with two sides polished. Firstly, after standard cleaning, $LiNbO_3$ wafers were immersed in molten benzoic acid at 230 ^oC for 4 h. A positive photoresist (AZ5214) was spin-coated onto the substrate. Stripes of 2-18 μ m width were patterned by photolithography. Then a metallic film with thickness of 150 nm was deposited on the samples by thermal evaporator (for Cr) or E-beam evaporator (for Ni and Ag). A Ti film of 10 nm was deposited before Ag deposition to enhance the adhesion between Ag and $LiNbO_3$. After lift-off, the plasma etching was performed in Plasma-Therm SLR770 ICP system. After ICP etching, metallic masks were removed by wet etching.

III. RESULTS AND DISCUSSION

Fig. 2 is a scanning electron microscope image which shows the surface condition of $LiNbO_3$ after ICP etching with 150nm Ag mask. The etching gas is CF_4/Ar mixture, and the etching time is 20 mins. It seems that the metallic mask was etched strongly with $LiNbO_3$ because there is no clear depth between the mask and the uncovered $LiNbO_3$ substrate. After ICP process, a Ag mask was hard to remove by strong acids such as HF or HCL+HNO_3. This is partly due to the reaction between Ag and etching gases. After etching, the composition of the metallic mask had already changed.

A similar situation happened when a Ni mask was used in ICP etching, as shown in Figs. 3(a) and (b). The gases we used were a SF_6/Ar mixture and CHF_3/Ar mixture. The surface roughness is obvious. Comparing with Ag, there is an advantage for Ni mask in that it can be easily removed by HNO_3. Fig. 3(a) shows the surface before removing the Ni mask. The surface roughness caused by physical sputtering and redeposition of the mask is clear in Fig. 3(a). Fig. 3(b) shows the surface after removing Ni mask.

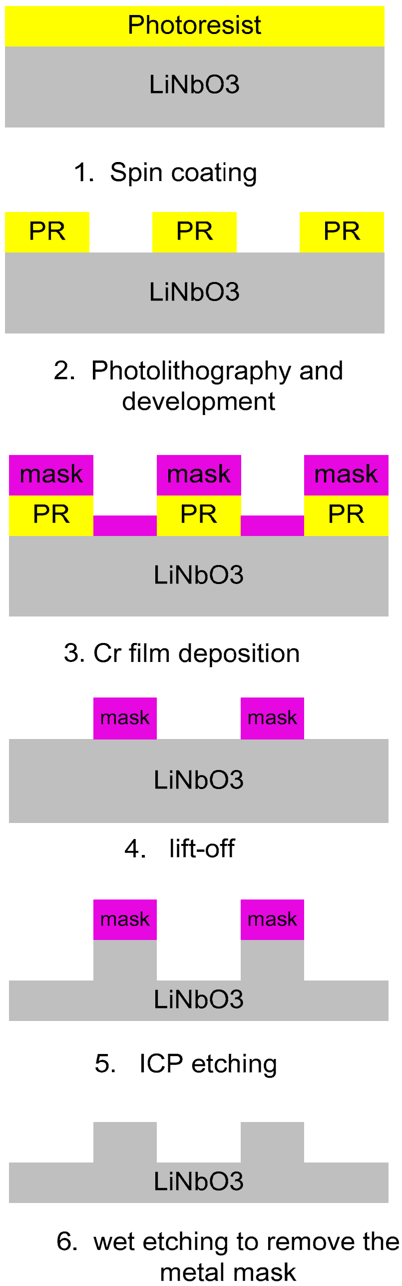


Fig. 1. Illustration of the ridge waveguide fabrication process

Fig. 4. (a) and (b) show the surface of $LiNbO_3$ after ICP and wet etching process using Cr as metal mask. By using CHF_3/Ar as etching gases, it can be found that the surface is very rough, especially in the uncovered area. It may be due to intensive ion bombardment to the mask and physical sputter. The Cr metal sputtered from the covered area is deposited on the uncovered area and serves as undesirable etch mask. The left part of Fig. 4(a) shows the details of the wavy morphology on the $LiNbO_3$ surface. The other important reason why CHF_3 and CF_4 cannot lead a smooth surface is LiF layer is formed during etching process (the structure of LiF layer is amorphous for CHF_3 while crystalline for CF_4) [10]. Comparing with use of CHF_3/Ar as an etching gas, SF_6/Ar combinations lead to better surface conditions and can avoid roughness greatly, as shown in Fig. 4(b). The etched area of the sample is very smooth. Cr mask can be easily removed by commercial etchant.

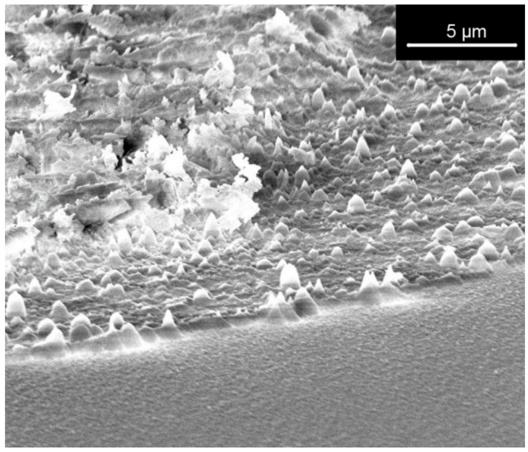
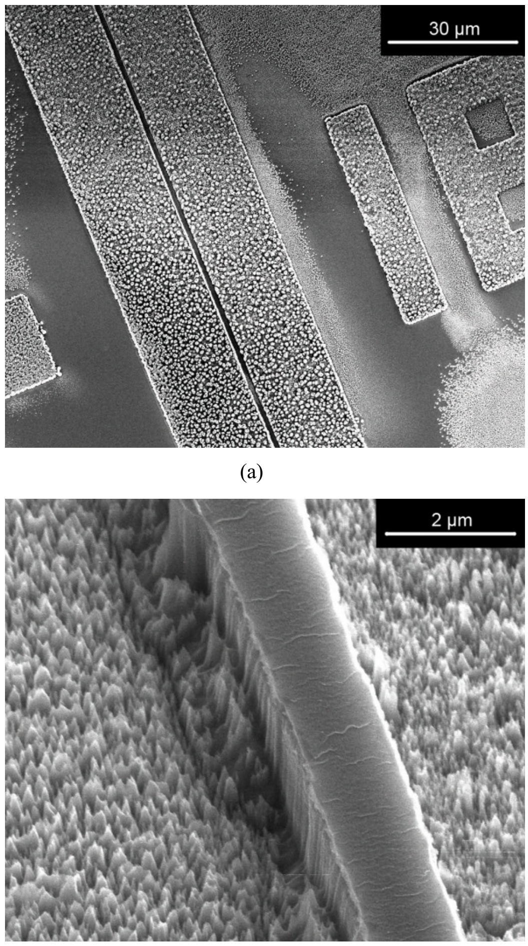


Fig. 2. SEM image of etched lithium niobate using Ag as etching mask. The $LiNbO_3$ substrate shows better surface condition than mask area.

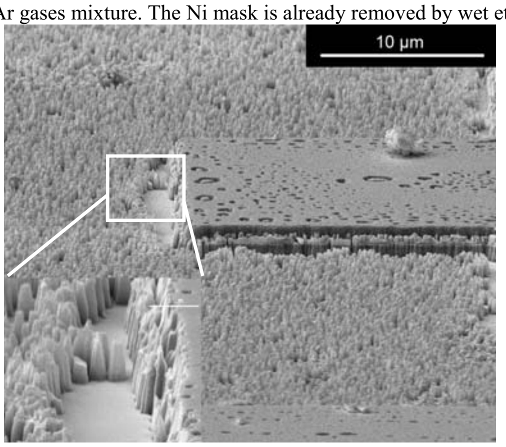


(a)

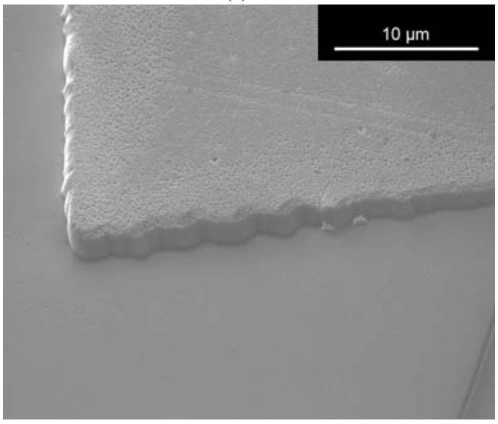
(b)

Fig. 3. SEM images of the surface condition of etched $LiNbO_3$ using Ni as etching mask. (a) Image of etched $LiNbO_3$ using SF_6/Ar gases mixture. The

Ni mask is remaining on the surface. (b) Image of etched $LiNbO_3$ using CHF_3/Ar gases mixture. The Ni mask is already removed by wet etching.



(a)



(b)

Fig. 4 (a) SEM image of etched $LiNbO_3$ using Cr mask and CHF_3/Ar gases mixture. The surface is very rough. The details of the uncovered region is showed in the left part of the image. (b) SEM image of etched $LiNbO_3$ using Cr mask and SF_6/Ar gases mixture.

Table I shows the ICP etching parameters, and the calculated etching rates and selectivities. Two masks, Cr and Ni were used because compared with Ag, they can be easily removed by commercial etchant. Three different gases were studied in ICP etching processes: CHF_3, CF_4 and SF_6. Argon is a useful inert gas for dry etching, which can used to physically bombard the target and enhance the sputtering process. In this work, Ar was added to the three gases to enhance the etching rate and profile. A surface profiler was used to measure etching depths shown in Table I.

In order to characterize the effect of different gases for ICP etching, the same etching parameters were used, gas pressure=8mTorr, ICP power=500W, RIE power=100W (No.1-4, Table I). All recipes show relatively high selectivity, ranged from 8.1:1 to 16:1. Recipe using Cr mask and CF_4/Ar mixture shows the highest selectivity, 16:1(No.3, Table I). A slightly small selectivity, 15.8:1, was also achieved in recipe 4 using Ni mask and CHF_3/Ar gases mixture. It means that both Cr and Ni are efficient etching masks for $LiNbO_3$-based device fabrication.

According to Table I (No.1-3), it can be seen that the etching rate of $LiNbO_3$ is dependent on the gas mixture. For Cr mask, the highest etching rate, 94nm/min, can be reached by using CHF_3/Ar gases (No.3). The cross section of the etched lithium niobate using CHF_3/Ar mixture and Recipe 3 is shown in Fig 5. A waveguide with depth of 1.95 μ m was obtained and the etching time is 20 mins. In addition, comparing No.1 and No.4, the etching rate of $LiNbO_3$ is found to be dependent on the etch mask. An etching rate of 79nm/min can be reached by using Ni mask; meanwhile, the etching rate of $LiNbO_3$ is only 57nm/min by using Cr mask under the same conditions.

Table I. Plasma etching parameters, average etching rate and selectivity

No	Mask	RIE (W)	ICP (W)	Temp. (°C)	Chamber Pressure (mTorr)	Gas (sccm)	Etching Time (min)	Etching Depth (nm)	Etching Rate (nm/min)	Selectivity
1	Cr	100	500	50	8.0	CHF_3: 25 Ar:20	20	1139	57	8.1:1
2	Cr	100	500	50	8.0	SF_6: 25 Ar: 20	20	950	48	12:1
3	Cr	100	500	50	8.0	CF_4:25 Ar: 20	20	1880	94	16:1
4	Ni	100	500	50	8.0	CHF_3: 25 Ar:20	20	1578	79	15.8:1
5	Cr	80	800	50	6.0	SF_6: 5 Ar: 5	10	476	48	9.6:1

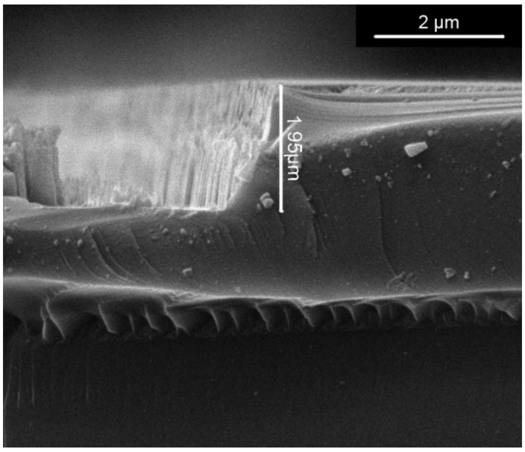


Fig. 5. SEM image of cross section of etched $LiNbO_3$. Etching gases: CHF_3/Ar; etching mask: Cr.

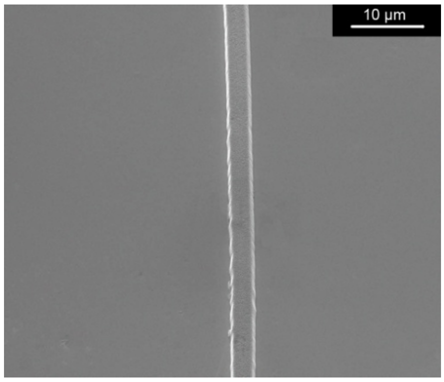
According to the comparison, we find that although CHF_3 or CF_4 can lead to higher etching rates, using a Cr mask combined with SF_6/Ar gases can lead to the best surface condition of etched $LiNbO_3$.Consequently, we fabricated ridge waveguides on $LiNbO_3$ substrate by using Cr mask combined with SF_6/Ar mixture. Fig. 6 (a) shows the top view of a ridge waveguide on $LiNbO_3$ fabricated by ICP etching using SF_6/Ar gases and Cr mask. In order to improve the profile of the waveguide, recipe 5 in Table I was used to fabricate the ridge waveguide with approximately 600nm depth, 5 μ m width, as shown in Fig. 6(b) and (c). Smooth top surface and nearly vertical sidewall were achieved.

IV. Conclusion

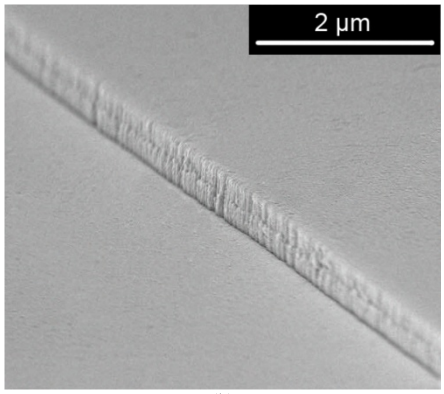
ICP etching characteristics of X-cut $LiNbO_3$ samples were studied under various metallic masks and etching conditions. By comparing different gases and masks, we find the roughness can be avoided greatly when using a Cr mask and SF_6 gas in the etching process. Mask selectivity > 15:1 was achieved under both Ni and Cr condition. A relatively high etch rate of 97.5 nm/min is achieved by using CHF_3/ Ar gases and a Cr mask. Using Ni mask and CHF_3/Ar gases also leads to a high etching rate: approximately 80nm/min. Optical ridge waveguides with approximately 600nm depth, smooth top surface and nearly vertical sidewalls were successfully fabricated using optimized etching conditions.

Acknowledgment

The authors acknowledge NUS Cross Faculty Grant R263000529646 in this work.



(a)



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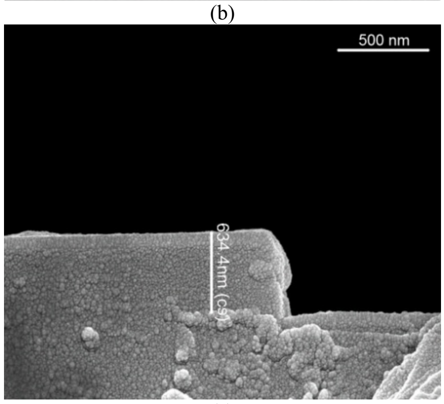

(c)

Fig. 6. (a) Top view of ridge waveguide on $LiNbO_3$ with smooth surface. (b)SEM image of ridge waveguide using recipe 5 in Table I. Surface roughness are greatly avoid. (c) SEM image showing cross section of the ridge waveguide.

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